Rev0 and Rev1-S Predicted Operating Points

AK-07 PCB coreless axial-flux motor — bench-test baseline — 19 October 2026 (rev. C)

These are **predicted operating points for planning the bench tests**, not continuous ratings. The continuous ratings will be set by testing.

The predictions come from the Maxwell simulations, the measured winding resistance (1.87–1.90 Ω on every phase of both stators) and the measured stator stack thickness (5.1 mm maximum). They supersede rev. B (19 October) and all earlier tables, including the 1.8 Ω / 0.85 A values in the 17 October technical report.

Changes from rev. B

- **Current.** 0.84 A pk per stator is now labelled a candidate thermal test point, not a continuous rating.
- **Speed.** 1000 rpm is the target test speed. The higher speeds are voltage limits only.
- **Rev1-S parallel.** Re-worked for the unequal back-EMF of S1 and S2.
- **Inductance.** Corrected from 110 µH to about 25 µH per stator, so the PWM ripple estimates are about 4.4 times higher.
- **Mechanical loss.** The method for measuring it has been corrected.
- **Drive.** The guidance is re-worded as checks.
- **Stack thickness.** Recorded as measured.

Predicted operating points at the 1000 rpm target test speed

All values are for **star (Y)** connection at 133.3 Hz electrical. The current is the **candidate thermal test point**, 0.84 A pk in any one stator. It has not been shown to keep the real winding below 60 °C (see "Thermal").

Quantity	Rev0	Rev1-S series	Rev1-S parallel
Supply for the test	12 V	24 V	12 V
Equivalent phase resistance (measured, about 20 °C)	1.89 Ω	3.78 Ω	0.945 Ω
Terminal (line) current	0.84 A pk (0.59 A rms)	0.84 A pk (0.59 A rms)	**1.54 A pk** (1.09 A rms)
Current in each stator	0.84 A pk	0.84 A pk	S1 0.84 / S2 0.70 A pk (unequal; see below)
Torque constant, terminal	27.8 mN·m/A pk	59.2 mN·m/A pk	29.5 mN·m/A pk (at this current)
Electromagnetic torque	23.4 mN·m	49.8 mN·m	45.5 mN·m
Estimated shaft torque[1]	18.2–21.2 mN·m	44.3–47.3 mN·m	40.0–43.0 mN·m
Shaft power	1.9–2.2 W	4.6–5.0 W	4.2–4.5 W
Back-EMF at 1000 rpm, phase / line	1.37 / 2.38 V rms	2.93 / 5.07 V rms	1.46 / 2.53 V rms[2]
Required DC bus, SVPWM (m = 0.95; 25–100 °C winding)	6.4–7.4 V	13.2–15.3 V	6.4–7.3 V
Copper loss (25–60 °C)	2.0–2.4 W	3.9–4.7 W	3.3–4.0 W
Electromagnetic efficiency, 60 °C	about 51 %	about 53 %	about 54 %
Calculated voltage-limited speed, 12 V[3]	2300–2590 rpm	570–840 rpm	2240–2490 rpm
Calculated voltage-limited speed, 24 V[3]	5690–5980 rpm	2160–2430 rpm	5420–5670 rpm

[1] Electromagnetic torque minus the predicted track-eddy drag (0.19 mN·m for Rev0, 0.42 mN·m for Rev1-S at 1000 rpm), minus an **assumed** 2–5 mN·m of mechanical loss. For example, 49.8 − 0.4 − (2 to 5) = 44.3–47.3 mN·m. The mechanical loss is not yet measured.

[2] Mean of S1 (1.37 V) and S2 (1.55 V) phase back-EMF. The terminal voltage of the parallel pair at no load is this mean.

[3] These are the speeds at which the inverter runs out of voltage at this current. **They are not speed ratings.** The rotor assembly, bearings and magnet retention have not been qualified for speeds above the 1000 rpm target. Do not exceed 1000 rpm until the mechanical assembly has been checked.

Rev1-S parallel connection

The simulations predict different torque constants for the two stators. S1 is 18.54 mN·m/A. S2 is 20.96 mN·m/A, higher because R1's steel backplate sits next to it. Their phase back-EMFs are therefore in phase but differ by about 13 % (1.37 against 1.55 V rms at 1000 rpm).

Connected in parallel, this difference drives a circulating current between S1 and S2:

- **At no load** it is about 0.06–0.07 A pk per phase at 1000 rpm, about 22–26 mW of extra loss.
- **Under load** it adds to S1's share and subtracts from S2's.

At 1.68 A pk terminal current, S1 would carry about 0.91 A pk and S2 0.77 A pk, so S1 would exceed the 0.84 A test point. The terminal current is therefore reduced to 1.54 A pk, which keeps S1 at 0.84 A pk. S1 is also the hotter stator in the Icepak model, so it sets the limit.

Terminal current alone cannot show either stator's current. If the parallel connection is used:

- measure the current of S1 and S2 separately (one current probe on each stator's phase A lead);
- confirm the back-EMF difference with the open-circuit test first.

Parallel remains an alternative that needs verification. Series connection on 24 V avoids the problem.

Connection

All values are for **star (Y)** connection. Each winding's start and finish are brought out (A1–A2, B1–B2, C1–C2), so delta is possible, but it is not covered here. In delta, the 5.5 % third-harmonic back-EMF drives about 40 mA rms circulating current at 1000 rpm.

The simulations excite the windings with balanced sinusoidal currents ($I_A + I_B + I_C = 0$), so they correspond directly to star.

Basis of each number

Input	Value	Status
Torque constant K (T = 1.5·K·I pk)	Rev0 18.53 mN·m/A. Rev1-S 39.50 mN·m/A (S1 18.54 + S2 20.96)	Simulated. Rev0 from the full-motor FEM. Rev1-S from the board-by-board sum, confirmed within 0.5 % by the Rev0-calibrated equivalent-coil model
Back-EMF	Phase peak = K·ω. THD 5.5 % phase, 0.7 % line	Simulated, rotating model
Phase resistance per stator	1.87–1.90 Ω, every phase of S1 and S2 (1.89 Ω used)	Measured. Temperature not recorded (15–30 °C assumed); 0–0.1 Ω treated as contact resistance
Resistance with temperature	1.85–1.96 Ω at 25 °C; 2.10–2.22 Ω at 60 °C; 2.38–2.52 Ω at 100 °C, per stator	Derived (α = 0.00393 /K)
Stator stack thickness	5.1 mm maximum	**Measured.** Matches the modelled 5.10 mm (5 × 1.02 mm), so stack spacing is no longer a significant uncertainty
Inductance	About 25 µH per stator (Rev1-S 50 µH series, 12.5 µH parallel)	Estimate from the Maxwell single-board value (≈ 2 µH) and the board-to-board coupling. Measure it before power-up
Mechanical loss	2–5 mN·m	**Assumed.** See “Measuring mechanical loss”
Thermal test point	0.84 A pk per stator	From an assumed 60 °C board limit and an assumed 15 K/W. Not established
Geometry	Backplate 86 / 19 mm, 1.0 mm air gaps	Model. Hardware backplates are 80 / 25 mm; air gaps not yet measured

Required voltage is E + I·R with the current on the q-axis. The winding reactance at 133 Hz is about 0.02 Ω, which is negligible. Each extra 0.25 mm of air gap lowers K and back-EMF by about 7 %, raising the current needed for a given torque.

Thermal

The 0.84 A pk test point comes from an assumed 15 K/W stack-to-air thermal resistance and an assumed 60 °C board limit. Neither is measured, so the test point does not establish that the real winding stays below 60 °C. The continuous rating must come from the heat run.

Thermal resistance, stack to air	60 °C board limit	100 °C board limit
8.75 K/W (earlier Icepak, hood with fans)	1.09 A pk per stator	1.51 A pk
15 K/W (pessimistic, used here)	0.84 A pk	1.15 A pk

- Run the heat test at constant current until the temperature settles. Hold 60 °C as the limit until the material of the yellow holders is confirmed. N35 magnets are limited to 80 °C.
- Log boards 1 and 5 with thermocouples. Take the mean winding temperature from the resistance rise: ΔT = (R/R0 − 1)/0.00393.
- The measured thermal resistance then sets the continuous current: I pk = √(ΔT_allowed / (1.5·R_hot·R_th)).
- Rev1-S stacks two stators closely, and S1 runs hotter than S2. Instrument S1.

Drive checks

With about 25 µH per stator, the winding inductance is too small to filter the PWM. Ripple is roughly V_dc / (6·L·f_sw), peak to peak. These estimates are uncertain by about ±50 %:

Winding	12 V, 20 kHz	12 V, 50 kHz	24 V, 50 kHz	12 V, 50 kHz, + 0.5 mH per phase
Rev0 (25 µH)	4.0 A	1.6 A	3.2 A	0.08 A
Rev1-S series (50 µH)	2.0 A	0.8 A	1.6 A	0.07 A
Rev1-S parallel (12.5 µH)	8.0 A	3.2 A	6.4 A	0.08 A

These are proposals to check against the available inverter, not an established drive setup:

1. Measure the phase inductance with an LCR meter (1–10 kHz) on each stator.
2. Check the inverter's switching frequency range, current-sense bandwidth and over-current trip against the table above.
3. At low current (0.2–0.3 A pk), measure the phase-current ripple with a current probe before increasing current.
4. If the ripple is more than about 20 % of the test current, raise the switching frequency if the inverter allows, or add series inductance of 0.5–1 mH per phase.
5. Set the driver's current limit on **peak** current, including ripple.

Measuring mechanical loss

A powered no-load input-power measurement cannot isolate mechanical loss: it also contains inverter loss, copper loss and eddy loss. Use one of these instead:

- **No-load current method (preferred with an FOC drive).** At a steady no-load speed, the electromagnetic torque equals the drag torque: $T_{drag} = 1.5 \cdot K \cdot I_{q,pk}$. Here K comes from the open-circuit back-EMF test ($K = E_{ph,pk} / \omega$) and I_q comes from the drive or a current probe. Subtract the predicted eddy drag (0.19 / 0.42 mN·m at 1000 rpm) to get the mechanical part. This excludes copper and inverter losses by construction.
- **Spin-down.** Spin the rotor to speed, disconnect it, and record speed against time. $T_{drag} = -J \cdot d\omega/dt$, which needs the rotor inertia J from CAD.
- **External drive with a torque sensor.** Drive the motor open-circuit from a second motor through an in-line torque sensor.

First-test sequence

1. **Measure before connecting.** Each stator's phase resistance, with room temperature noted. The phase inductance. The assembled air gaps.
2. **Open-circuit back-EMF (main model check).** Spin each motor externally at known speeds up to 1000 rpm and record the line back-EMF.
   - Expected at 1000 rpm: Rev0 2.38 V rms; Rev1-S 5.07 V rms (series).
   - For Rev1-S, also record S1 and S2 separately: about 2.38 and 2.69 V rms line. This confirms they are in phase and quantifies their difference before any parallel connection.
   - A value more than a few percent low points to a larger air gap than modelled.
3. **First powered run.** Power Rev0 at 0.2–0.3 A pk and 300–500 rpm and check the current ripple (Drive checks). Then step up towards 0.84 A pk at 1000 rpm, logging temperature.
4. **Mechanical loss.** Measure the no-load drag with one of the methods above, then torque against current.
5. **Heat run.** Run at constant current until temperatures settle, to measure the thermal resistance and set the continuous rating.
6. **Rev1-S.** Repeat in series. Use parallel only with per-stator current measurement.